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COMMENT

Hypertensive heart disease: is it really a pathology?

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Hypertensive heart disease (HHD) is a diagnosis encompassing cardiac structural complications attributed to systemic hypertension.

The term hypertensive heart disease (HHD) appears as a formal diagnosis in the International Classification of Diseases, 10th Revision (ICD-10), under codes I11.0 (with heart failure) and I11.9 (without heart failure). These codes imply a causal relationship between systemic hypertension and cardiac structural or functional pathology, potentially culminating in heart failure. As such, the term HHD suggests the existence of a definable disease process, presumably progressive and clinically significant.

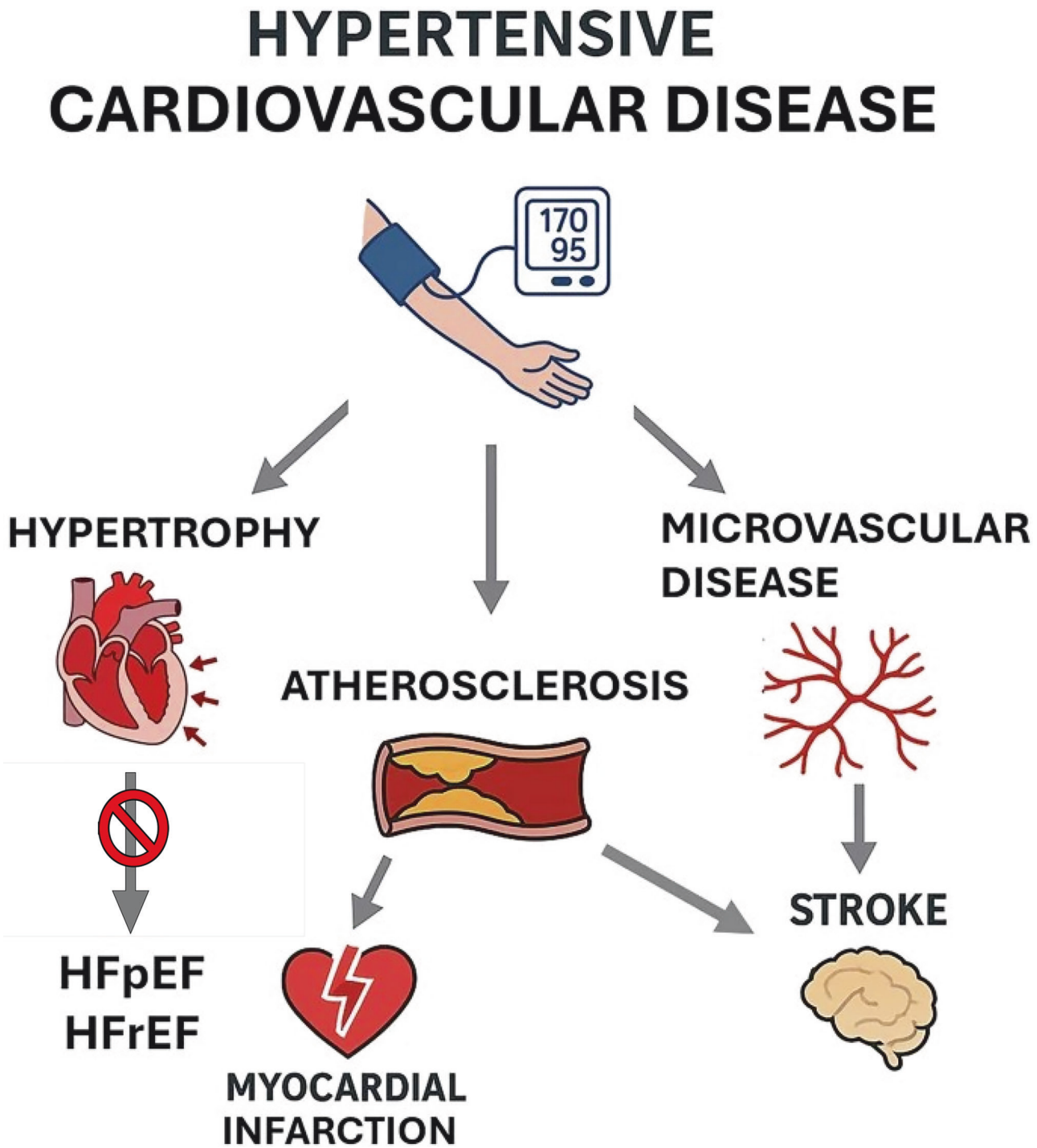
However, the use of HHD in real-world cardiology fails to correspond to a clinically recognizable syndrome. In clinical practice, the label is frequently applied to hypertensive patients who show any combination of left ventricular hypertrophy (LVH), diastolic dysfunction, or left atrial enlargement—echocardiographic features that are common and often paraphysiological with aging. Notably, left ventricular wall thickening can lead to heart failure only when it arises from a replacement of muscular myocardial tissue, as seen in infiltrative diseases such as amyloidosis or storage disorders like Fabry disease. In contrast, true muscular hypertrophy, that is the cause of cardiac wall tickening in hypertension, is rarely associated with heart decompensation even in cases of pathologic degrees of thickening, such as in sarcomeric hypertrophic cardiomyopathies [1]. Indeed in this pathology haemodynamic deterioration typically results from mechanical complications—most notably left ventricular outflow tract (LVOT) obstruction and severe mitral regurgitation due to systolic anterior motion (SAM) of the mitral valve-- not diastolic dysfunction due to cardiac wall hypertrophy [2, 3]. Moreover, these patients almost always maintain preserved left ventricular systolic function. Similarly, in hypertensive patients, systolic dysfunction is exceedingly rare and usually only seen in cases of prolonged, uncontrolled severe hypertension, where transient myocardial stunning may occur—essentially a reversible form of strain-related dysfunction rather than a progressive heart failure syndrome. Thus, myocardial muscular hypertrophy per se, even when severe, is not sufficient to cause heart failure. With regard to diastolic left ventricular filling, grade I diastolic dysfunction is the most frequently observed pattern in hypertensive patients and indicates normal left ventricular dilling pressure, instead higher grades of diastolic dysfunction are almost never reached in hypertensive heart. Grade I dyastolic dysfunction is commonly see also in older individuals and is generally considered a paraphysiological adaptation to aging rather than a sign of pathology. Similarly, left atrial enlargement tends to frequently develop with age as a paraphysiological adaptation and it is not linked with heart decompensation, except in some rare patient with permanent atrial fibrillation in which severe atrial enlargement may cause severe mitral and tricuspid valve regurgitation due to annular dilatation. Instead, mild or moderate atrial enlargement (and also most cases of severe enlargement) are neither specific to hypertension nor reliably predictive of adverse outcomes. While left atrial enlargement can be associated with increased cardiovascular risk in some populations, its prognostic significance is limited when it occurs in isolation, particularly in older individuals without accompanying structural or electrical abnormalities [4]. There is no combination of left ventricular hypertrophy, atrial enlargement, and grade I diastolic dysfunction—features commonly observed in hypertension and aging—that independently leads to heart failure. These findings are paraphysiologic and asymptomatic [5, 6], lacking the symptomatic or progressive features typical of recognized

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Graphical Opinion

Hypertensive heart disease is not a direct cause of heart failure. Left ventricular hypertrophy is common in hypertension but rarely progresses to HFpEF or HFrEF. The true cardiovascular consequences of hypertension are primarily mediated by atherosclerosis and microvascular dysfunction, leading to stroke and myocardial infarction.



disease entities. While they may be associated with an increased cardiovascular risk, for a condition to be considered a disease entity, it should present with recognizable symptoms, follow a relatively defined pathophysiological process, and be associated with a predictable clinical course. In the case of so-called HHD, no such course exists because

eft ventricular hypertrophy, when present, acts merely as a surrogate marker of overall cardiovascular load or hypertension severity and duration, but it does not follow a defined or autonomous trajectory toward heart failure. Instead, pathophysiological mechanisms through which hypertension affects the vascular system—such as vascular remodeling and aterosclerosis—occur at a different systemic level.

However, hypertensive patients do have a higher risk of adverse cardiovascular outcomes compared to normotensive individuals. Crucially, these outcomes are not related to the development of heart failure (hypertensive heart does not develop heart failure), but rather to atherosclerotic events—particularly myocardial infarction and stroke, with the latter also associated with the development of atrial fibrillation in hypertensive individuals and microvascular dysfunction [7, 8]. Despite this, routine cardiology follow-up often involves annual echocardiograms, aimed at detecting heart failure—especially HFpEF—while coronary artery disease is frequently overlooked unless symptoms are overt. This surveillance paradox is striking: we routinely search for a poorly defined entity (HFpEF), whose very pathophysiological existence is debated and even refuted by recent literature [9, 10], while ignoring ischemic disease, which is both prevalent and prognostically significant in hypertension. Moreover, chronic critical coronary stenoses are often asymptomatic, with myocardial infarction being the first clinical manifestation in many cases [9].

Because of its inclusion in ICD coding systems, HHD term is often used in clinical records, hospital billing, and epidemiological databases. This creates the illusion that HHD is a widely prevalent pathology that leads inexorably to myocardial tissue damage, contributing to an obsessive focus on blood pressure and proliferation of subcategories such as ‘morning hypertension’ or ‘digital hypertension’.

Echocardiography frequently plays a role in perpetuating the label. Slight increases in wall thickness or left atrial volume are often described as “signs of hypertensive heart disease”. When these findings are paired with common symptoms such as fatigue or exertional dyspnea, clinicians may infer a pathophysiological link—despite the absence of objective evidence of cardiac malfunction.

This circular reasoning—hypertension plus nonspecific echo findings equals HHD—contributes to making patients feel seriously ill without justification, while prompting healthcare systems to invest in frequent and costly follow-up investigations of questionable value. Modern cardiology has embraced precision: ischemic, valvular, and genetic cardiomyopathies are each characterized by specific mechanisms and clinical trajectories. HHD, by contrast, remains a diffuse, nonspecific construct.

Rather than using a label with no clear diagnostic boundaries or prognostic implications, clinicians should describe what they actually see: a within limit echooardiogram in a hypertensive patients. This description is more accurate and informative than invoking a broad term like HHD, which conveys little and implies much.

From a scientific standpoint, retiring the label—or strictly redefining it based on robust, reproducible criteria—would improve both clinical care and research integrity. Hypertensive heart disease is a diagnostic label born of coding systems, not clinical necessity. It does not describe a specific syndrome, nor does it reflect a consistent disease trajectory. As currently used, it serves more to justify documentation than to guide care.

In the interest of precision medicine, we propose that the term be abandoned in favor of explicit, evidence-based descriptions of cardiac structure and function.

Compliance with ethical standards

Conflict of interest The author declares no competing interests.

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